Practice workbook · Lengths in a 3D solid using right-angled triangles

Finding diagonals, heights and sloping edges of pyramids, prisms and cuboids by using Pythagoras in one or two right-angled triangles.

22 question parts: 1 worked-example retry and 21 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-13

1. Question 11 · part a · 3 marks

2025 NOV · Paper 02

Worked-example retry · Attempt part a only.

11

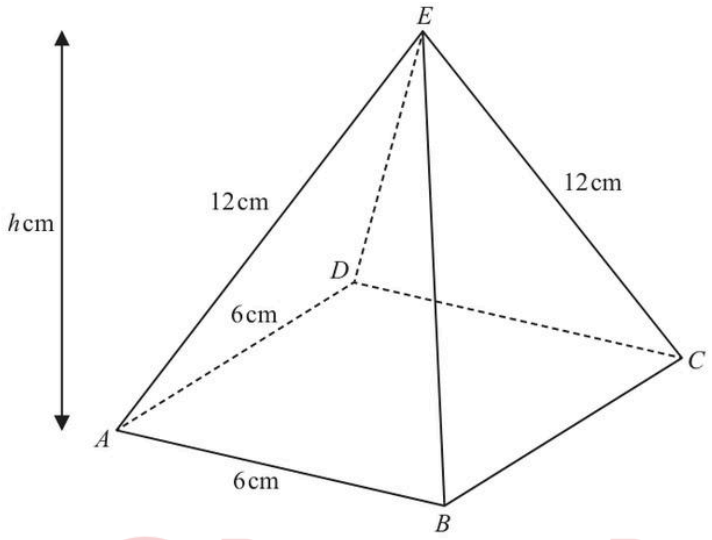


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a right pyramid $ABCDE$ with

- vertex E
- horizontal square base $ABCD$ of side 6 cm
- $EA = EB = EC = ED = 12$ cm
- vertical height of h cm

(a) Show that $h = 3\sqrt{14}$

(3)

(b) Find, in degrees to one decimal place, the size of the angle between EB and the plane $ABCD$

(2)

The midpoint of AE is P and the midpoint of BE is Q

(c) Find, to the nearest degree, the size of the obtuse angle between the plane EPQ and the plane $PQCD$

(6)

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1. Question 11 · part a · 3 marks · Working and answer

Working and answer

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2. Question 3 · part a · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part a only.

3

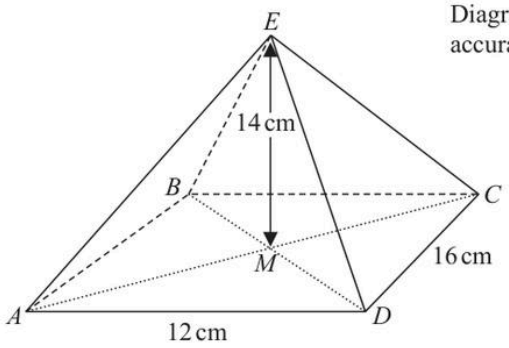


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the right pyramid $ABCDE$. The base, $ABCD$, of the pyramid is a horizontal rectangle with $AD = 12$ cm and $CD = 16$ cm. The height ME of the pyramid is 14 cm where M is the point of intersection of the diagonals of the base.

The sloping edges, EA, EB, EC and ED of the pyramid are all of equal length.

(a) Calculate, to 3 significant figures, the length of a sloping edge. **(3)**

Calculate, in degrees to one decimal place, the size of

(b) the angle between AE and the base, **(3)**

(c) the angle between the plane AED and the base. **(3)**

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P 6 2 2 8 0 A 0 6 3 2

2. Question 3 · part a · 3 marks · Working and answer

Working and answer

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3. Question 8 · part b · 3 marks

2025 JUN · Paper 01

Further practice · Attempt part b only.

8

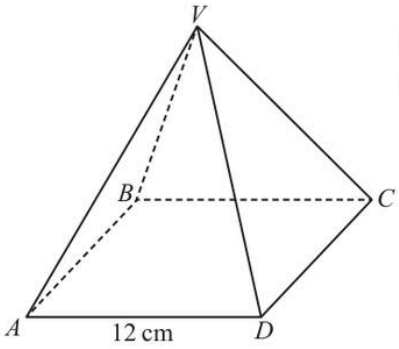


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a right pyramid with vertex V and square base, $ABCD$, of side 12 cm

The size of the angle between the plane VAB and the base $ABCD$ is $30°$

(a) Show that the height of the pyramid is $2\sqrt{3}$ cm

(2)

(b) Find, in cm, the exact length of VA

(3)

(c) Find, in cm^2, the exact area of triangle VAD

(3)

(d) Find, in cm, the exact length of the perpendicular from D to VA

(2)

(e) Find, in degrees to one decimal place, the size of the obtuse angle between the plane VAB and the plane VAD

(3)

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3. Question 8 · part b · 3 marks · Working and answer

Working and answer

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4. Question 8 · part b · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

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Diagram **NOT** accurately drawn

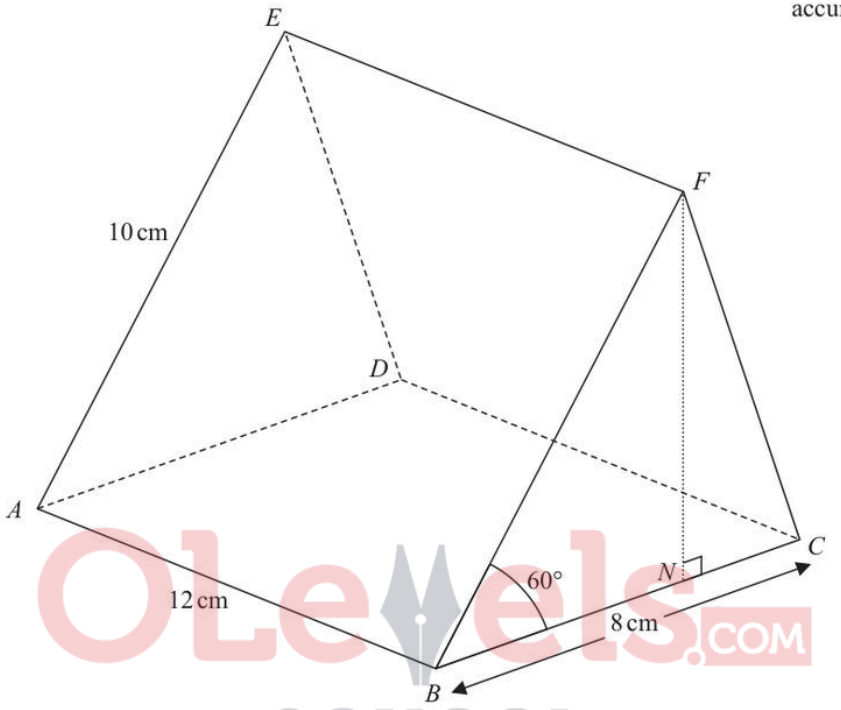


Figure 3

Figure 3 shows a right prism $ABCDEF$. The cross section BCF of the prism is a triangle.

$AB = DC = 12\,\text{cm}$ $BC = AD = 8\,\text{cm}$ $BF = AE = 10\,\text{cm}$ $\angle FBC = \angle EAD = 60°$

The point N lies on BC such that FN is perpendicular to BC.

(a) Show that $BN = 5\,\text{cm}$.

(2)

(b) Find, in cm to 3 significant figures, the length of EN.

(3)

The midpoint of BF is X and the midpoint of FC is Y.

(c) Find, in degrees to one decimal place, the size of the angle between the plane $ABCD$ and the plane $AXYD$.

(2)

(d) Find, in degrees to one decimal place, the size of the angle AYE.

(6)

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P 6 1 8 8 3 A 0 2 2 3 6

4. Question 8 · part b · 3 marks · Working and answer

Working and answer

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5. Question 9 · part b · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

9 Given that α is the acute angle such that $\tan\alpha = \frac{2}{3}$

(a) find the exact value of $\cos\alpha$

(1)

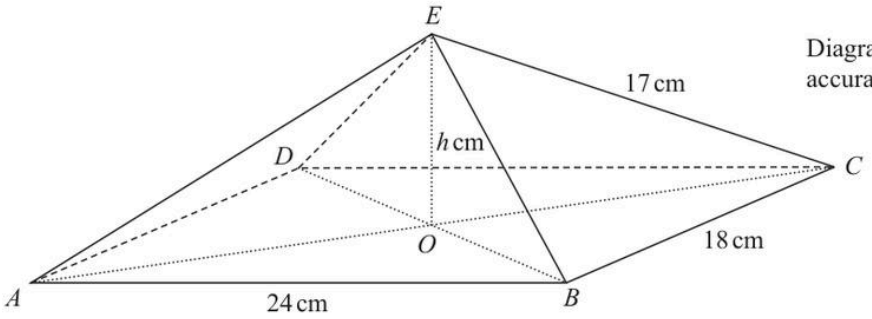


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a right pyramid with a rectangular base $ABCD$ and vertex E

The rectangular base of the pyramid is horizontal with $AB = 24$ cm and $BC = 18$ cm.

The diagonals of the base intersect at the point O

The vertex E of the pyramid is vertically above O such that

$$AE = BE = CE = DE = 17 \text{ cm}$$

The height of the pyramid is h cm.

(b) Find the value of h

(3)

The size of the angle between the plane EBC and the plane $ABCD$ is $\theta°$

(c) Show that $\tan\theta° = \frac{2}{3}$

(2)

The point P is the midpoint of EB and the point Q is the midpoint of EC

(d) Find the size, in degrees to one decimal place, of the angle between the plane OPQ and the plane $BCQP$

(4)

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5. Question 9 · part b · 3 marks · Working and answer

Working and answer

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6. Question 11 · part b · 3 marks

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Further practice · Attempt part b only.

11

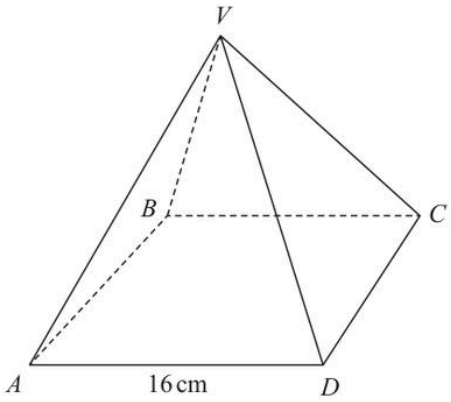


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a right pyramid with vertex V and square base, $ABCD$, of side 16 cm.

The size of angle AVC is 90°

(a) Show that the height of the pyramid is $8\sqrt{2}$ cm.

(4)

(b) Find, in cm, the length of VA.

(3)

(c) Find, in cm, the exact length of the perpendicular from D onto VA.

(3)

Find, in degrees to one decimal place, the size of

(d) the angle between the plane VAB and the base $ABCD$,

(3)

(e) the obtuse angle between the plane VAB and the plane VAD.

(3)

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6. Question 11 · part b · 3 marks · Working and answer

Working and answer

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7. Question 4 · part a · 2 marks

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Further practice · Attempt part a only.

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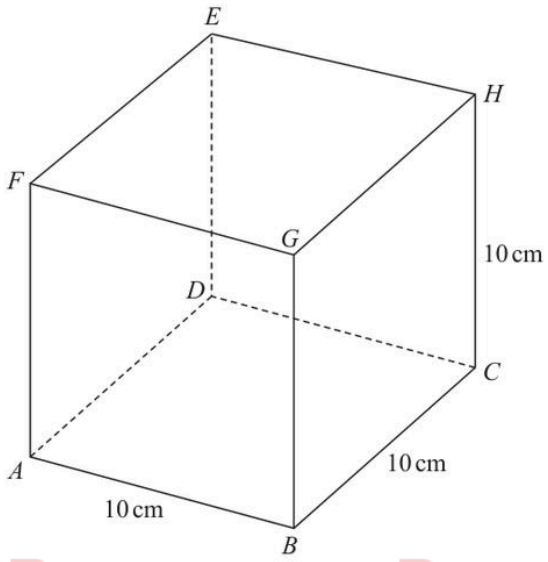


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a solid cube $ABCDEFGH$ with sides of length 10 cm.

(a) Find, in cm, the exact length of FC

(2)

(b) Find, in degrees to one decimal place, the size of $\angle FCA$

(2)

(c) Find, to the nearest degree, the size of the obtuse angle between the plane BFH and the plane BHC

(5)

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7. Question 4 · part a · 2 marks · Working and answer

Working and answer

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8. Question 3 · part a · 2 marks

2020 JAN · Paper 02

Further practice · Attempt part a only.

3

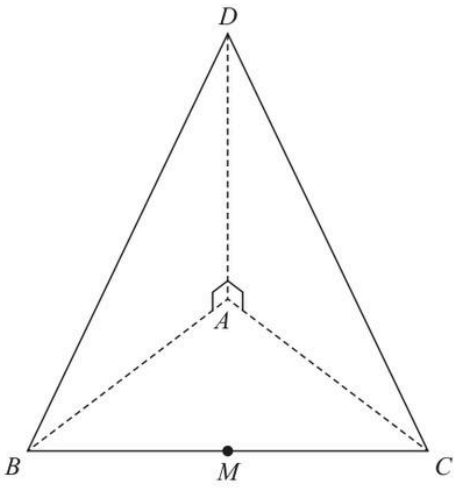


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a triangular pyramid $ABCD$.

The base, ABC, of the pyramid is a horizontal isosceles triangle with $AB = AC = 10\,\text{cm}$ and $BC = 16\,\text{cm}$. The midpoint of BC is M.

The face BCD of the pyramid is an isosceles triangle with $BD = CD = 26\,\text{cm}$ and D is vertically above A.

$\angle BAD = \angle CAD = 90°$

(a) Calculate the length, in cm, of AM.

(2)

Calculate, in degrees to the nearest degree,

(b) the size of $\angle BCD$,

(3)

(c) the size of the angle between the planes BCA and BCD.

(4)

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8. Question 3 · part a · 2 marks · Working and answer

Working and answer

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9. Question 9 · part a · 2 marks

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Further practice · Attempt part a only.

9

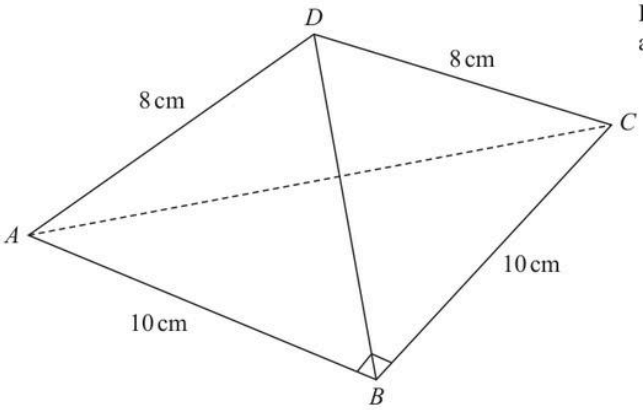


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a triangular pyramid $ABCD$ with base ABC

$AB = BC = 10$ cm $\quad AD = CD = 8$ cm $\quad \angle ABC = 90°$

(a) Find the exact length of AC

Give your answer in the form $p\sqrt{q}$ cm where p is an integer and q is a prime number.

(2)

The point M is the midpoint of AC

(b) Find the exact length of BM

Give your answer in the form $m\sqrt{n}$ cm where both m and n are prime numbers.

(2)

Given that $BD = 6$ cm,

(c) find, in degrees to one decimal place, the size of the acute angle between the plane ACD and the plane ABC

(4)

The base ABC of the pyramid is placed on a horizontal plane.

(d) Find, in cm to 3 significant figures, the vertical height of D above the base.

(2)

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9. Question 9 · part a · 2 marks · Working and answer

Working and answer

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10. Question 9 · part b · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

9

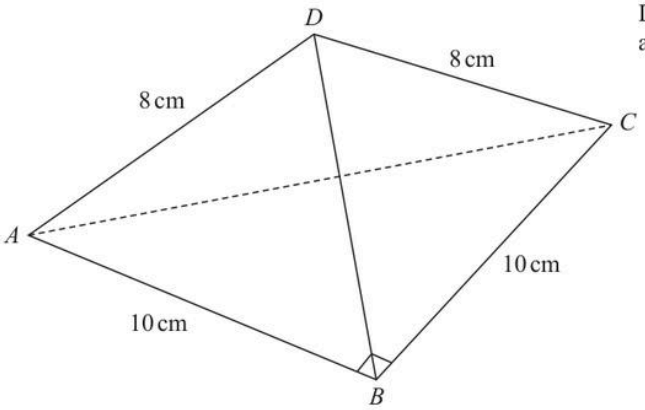


Diagram **NOT** accurately drawn

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Figure 2 shows a triangular pyramid $ABCD$ with base ABC

$AB = BC = 10$ cm $\quad AD = CD = 8$ cm $\quad \angle ABC = 90°$

(a) Find the exact length of AC

Give your answer in the form $p\sqrt{q}$ cm where p is an integer and q is a prime number.

(2)

The point M is the midpoint of AC

(b) Find the exact length of BM

Give your answer in the form $m\sqrt{n}$ cm where both m and n are prime numbers.

(2)

Given that $BD = 6$ cm,

(c) find, in degrees to one decimal place, the size of the acute angle between the plane ACD and the plane ABC

(4)

The base ABC of the pyramid is placed on a horizontal plane.

(d) Find, in cm to 3 significant figures, the vertical height of D above the base.

(2)

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10. Question 9 · part b · 2 marks · Working and answer

Working and answer

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11. Question 9 · part a · 2 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

9

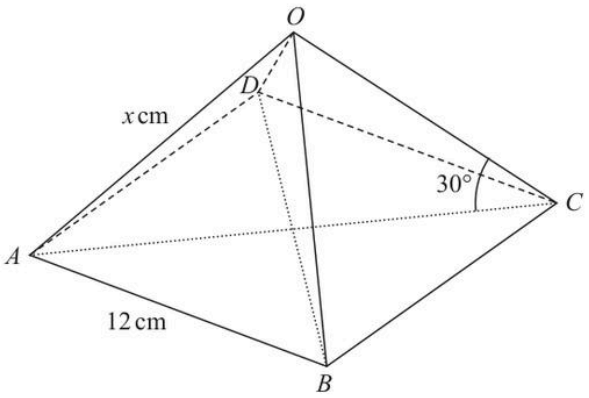


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows the right pyramid $OABCD$ with a square base $ABCD$ of side 12 cm.

$OA = OB = OC = OD = x$ cm and $\angle OAC = \angle ODB = \angle OCA = \angle OBD = 30°$

(a) Find the exact length of AC

(2)

(b) Show that $x = 4\sqrt{6}$

(2)

(c) Find the total surface area, to the nearest cm^2, of the pyramid.

(5)

(d) Find the size of the obtuse angle, to the nearest degree, between the plane OAB and the plane OBC

(4)

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11. Question 9 · part a · 2 marks · Working and answer

Working and answer

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12. Question 6 · part a · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part a only.

6

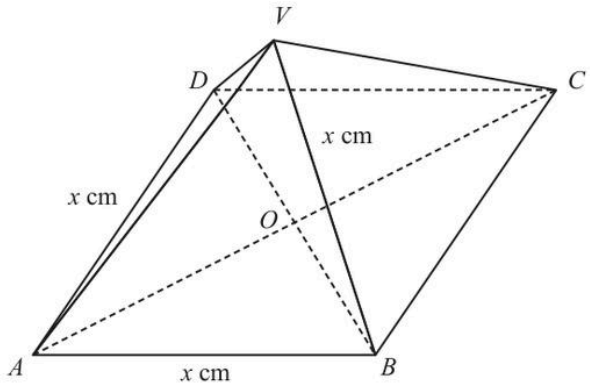


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a right pyramid with a horizontal square base.

$$AB = BC = CD = DA = x \text{ cm}$$

$$AV = BV = CV = DV = x \text{ cm}$$

O is the point of intersection of the diagonals of the base.
The vertex V of the pyramid is vertically above O

(a) Show that $VO = \frac{\sqrt{2}}{2}x$ cm

(3)

(b) Find, in degrees, the size of the angle AVC

(2)

(c) Find, in degrees to one decimal place, the size of the angle between the plane VAB and the plane VDC

(3)

The volume of the pyramid is 200 cm^3

Given that the volume of a pyramid $= \frac{1}{3} \times$ base area $\times$ height

(d) Find to 3 signficant figures, the value of x

(3)

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P 7 3 5 8 4 A 0 1 6 3 6

12. Question 6 · part a · 3 marks · Working and answer

Working and answer

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13. Question 8 · part a · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part a only.

8

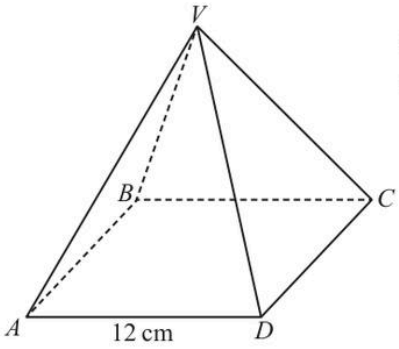


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a right pyramid with vertex V and square base, $ABCD$, of side 12 cm

The size of the angle between the plane VAB and the base $ABCD$ is $30°$

(a) Show that the height of the pyramid is $2\sqrt{3}$ cm

(2)

(b) Find, in cm, the exact length of VA

(3)

(c) Find, in cm^2, the exact area of triangle VAD

(3)

(d) Find, in cm, the exact length of the perpendicular from D to VA

(2)

(e) Find, in degrees to one decimal place, the size of the obtuse angle between the plane VAB and the plane VAD

(3)

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13. Question 8 · part a · 2 marks · Working and answer

Working and answer

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14. Question 8 · part a · 2 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

8

Diagram **NOT** accurately drawn

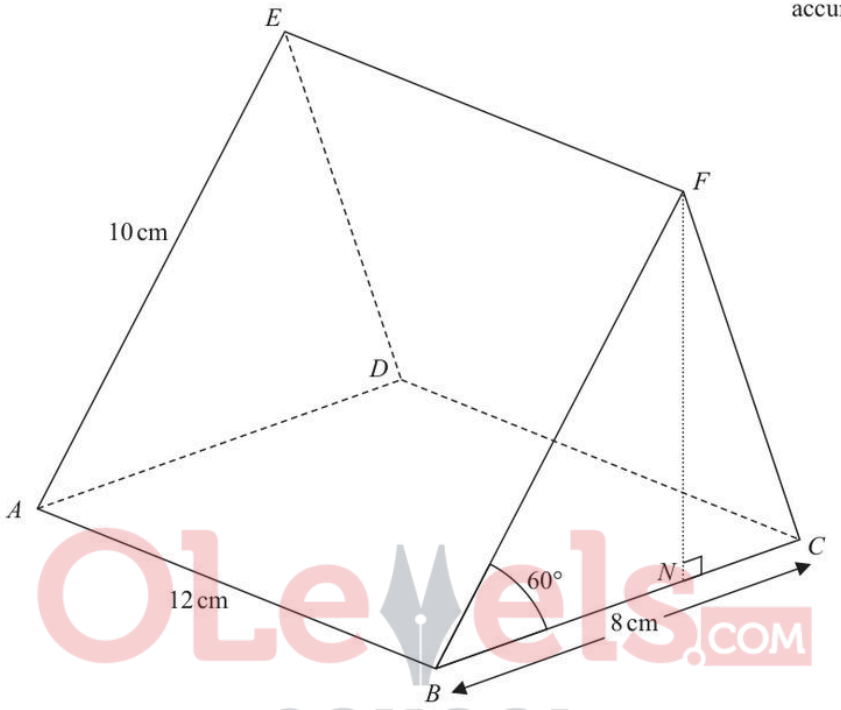


Figure 3

Figure 3 shows a right prism $ABCDEF$. The cross section BCF of the prism is a triangle.

$AB = DC = 12\,\text{cm}$ $BC = AD = 8\,\text{cm}$ $BF = AE = 10\,\text{cm}$ $\angle FBC = \angle EAD = 60°$

The point N lies on BC such that FN is perpendicular to BC.

(a) Show that $BN = 5\,\text{cm}$.

(2)

(b) Find, in cm to 3 significant figures, the length of EN.

(3)

The midpoint of BF is X and the midpoint of FC is Y.

(c) Find, in degrees to one decimal place, the size of the angle between the plane $ABCD$ and the plane $AXYD$.

(2)

(d) Find, in degrees to one decimal place, the size of the angle AYE.

(6)

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14. Question 8 · part a · 2 marks · Working and answer

Working and answer

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15. Question 9 · part a · 1 mark

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

9 Given that α is the acute angle such that $\tan\alpha = \frac{2}{3}$

(a) find the exact value of $\cos\alpha$

(1)

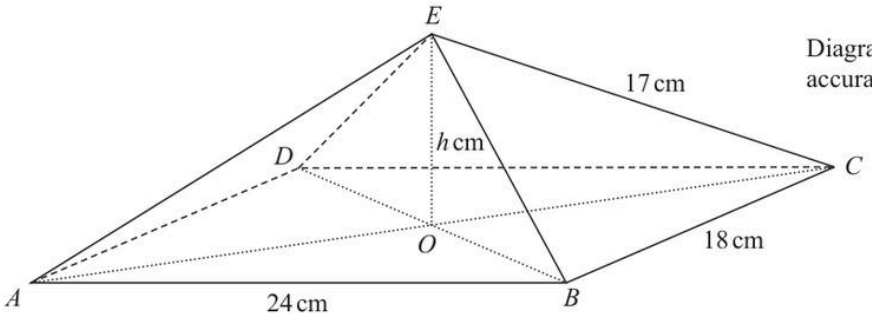


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a right pyramid with a rectangular base $ABCD$ and vertex E

The rectangular base of the pyramid is horizontal with $AB = 24$ cm and $BC = 18$ cm.

The diagonals of the base intersect at the point O

The vertex E of the pyramid is vertically above O such that

$$AE = BE = CE = DE = 17 \text{ cm}$$

The height of the pyramid is h cm.

(b) Find the value of h

(3)

The size of the angle between the plane EBC and the plane $ABCD$ is $\theta°$

(c) Show that $\tan\theta° = \frac{2}{3}$

(2)

The point P is the midpoint of EB and the point Q is the midpoint of EC

(d) Find the size, in degrees to one decimal place, of the angle between the plane OPQ and the plane $BCQP$

(4)

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P 6 6 3 0 7 A 0 2 2 3 2

15. Question 9 · part a · 1 mark · Working and answer

Working and answer

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16. Question 11 · part a · 4 marks

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Further practice · Attempt part a only.

11

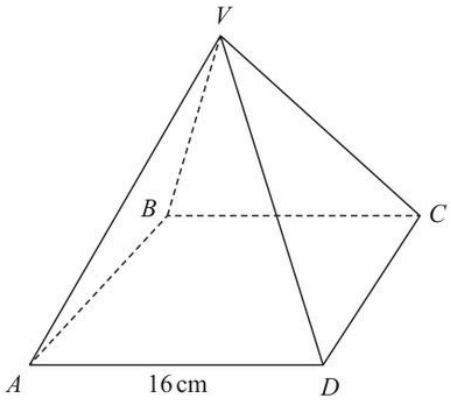


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a right pyramid with vertex V and square base, $ABCD$, of side 16 cm.

The size of angle AVC is 90°

(a) Show that the height of the pyramid is $8\sqrt{2}$ cm.

(4)

(b) Find, in cm, the length of VA.

(3)

(c) Find, in cm, the exact length of the perpendicular from D onto VA.

(3)

Find, in degrees to one decimal place, the size of

(d) the angle between the plane VAB and the base $ABCD$,

(3)

(e) the obtuse angle between the plane VAB and the plane VAD.

(3)

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16. Question 11 · part a · 4 marks · Working and answer

Working and answer

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17. Question 5 · part a · 3 marks

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Further practice · Attempt part a only.

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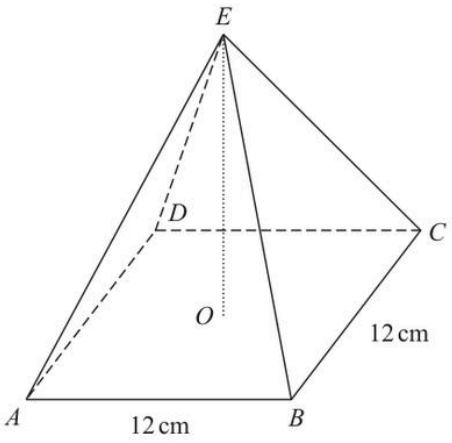


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a right pyramid with a square base $ABCD$ and vertex E.

The base of the pyramid is horizontal with $AB = BC = 12$ cm.
The diagonals of the base intersect at the point O.

The vertex E of the pyramid is vertically above O and the angle between EA and the plane $ABCD$ is 30°

The height of the pyramid is h cm.

(a) Find the exact value of h

(3)

The point F lies on AD such that $AF : FD = 1 : 4$

(b) Calculate, to the nearest degree, the size of the angle EFO.

(4)

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17. Question 5 · part a · 3 marks · Working and answer

Working and answer

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18. Question 10 · part a · 3 marks

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Further practice · Attempt part a only.

10

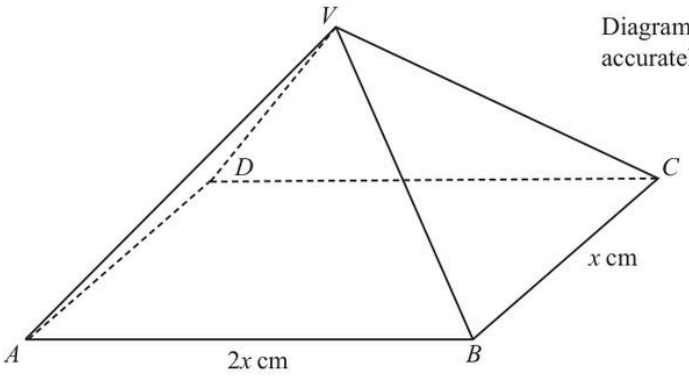


Figure 4

Figure 4 shows a right pyramid $ABCDV$

The base of the pyramid is a rectangle where,

$AB = DC = 2x$ cm $AD = BC = x$ cm

The edges VA, VB, VC and VD are all of equal length.

The angle between VA and $ABCD$ is 45°

(a) Show that $VA = \frac{\sqrt{10}}{2}x$ cm

(3)

(b) Find in cm, the exact height of the pyramid in terms of x

(2)

Find, in degrees to one decimal place,

(c) the size of angle VBA

(2)

(d) the size of the obtuse angle between the plane AVC and the plane BVD

(4)

Given that the volume of the pyramid is $9\sqrt{5}$ cm^3

(e) find the value of x

(2)

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18. Question 10 · part a · 3 marks · Working and answer

Working and answer

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19. Question 10 · part b · 2 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

10

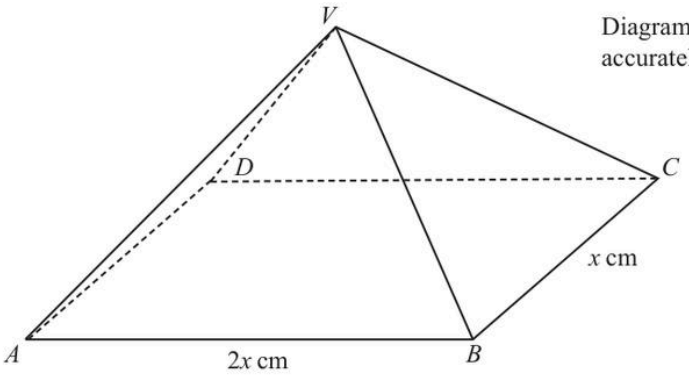


Figure 4

Figure 4 shows a right pyramid $ABCDV$

The base of the pyramid is a rectangle where,

$AB = DC = 2x$ cm $AD = BC = x$ cm

The edges VA, VB, VC and VD are all of equal length.

The angle between VA and $ABCD$ is 45°

(a) Show that $VA = \frac{\sqrt{10}}{2}x$ cm

(3)

(b) Find in cm, the exact height of the pyramid in terms of x

(2)

Find, in degrees to one decimal place,

(c) the size of angle VBA

(2)

(d) the size of the obtuse angle between the plane AVC and the plane BVD

(4)

Given that the volume of the pyramid is $9\sqrt{5}$ cm^3

(e) find the value of x

(2)

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19. Question 10 · part b · 2 marks · Working and answer

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20. Question 9 · part b · 2 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

9

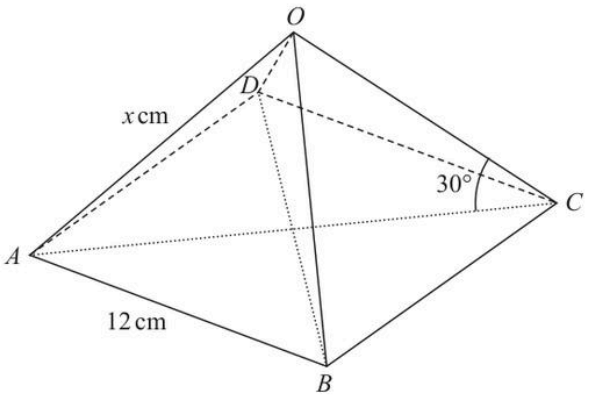


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows the right pyramid $OABCD$ with a square base $ABCD$ of side 12 cm.

$OA = OB = OC = OD = x$ cm and $\angle OAC = \angle ODB = \angle OCA = \angle OBD = 30°$

(a) Find the exact length of AC

(2)

(b) Show that $x = 4\sqrt{6}$

(2)

(c) Find the total surface area, to the nearest cm^2, of the pyramid.

(5)

(d) Find the size of the obtuse angle, to the nearest degree, between the plane OAB and the plane OBC

(4)

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20. Question 9 · part b · 2 marks · Working and answer

Working and answer

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21. Question 9 · part d · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part d only.

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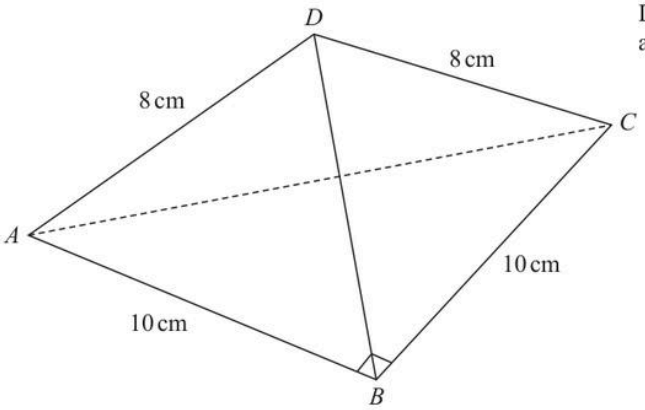


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a triangular pyramid $ABCD$ with base ABC

$AB = BC = 10$ cm $\quad AD = CD = 8$ cm $\quad \angle ABC = 90°$

(a) Find the exact length of AC

Give your answer in the form $p\sqrt{q}$ cm where p is an integer and q is a prime number.

(2)

The point M is the midpoint of AC

(b) Find the exact length of BM

Give your answer in the form $m\sqrt{n}$ cm where both m and n are prime numbers.

(2)

Given that $BD = 6$ cm,

(c) find, in degrees to one decimal place, the size of the acute angle between the plane ACD and the plane ABC

(4)

The base ABC of the pyramid is placed on a horizontal plane.

(d) Find, in cm to 3 significant figures, the vertical height of D above the base.

(2)

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P 6 6 0 2 7 R R A 0 2 6 3 6

21. Question 9 · part d · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part d only.

Question 9 continued

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21. Question 9 · part d · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part d only.

Question 9 continued

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28

P 6 6 0 2 7 R R A 0 2 8 3 6

21. Question 9 · part d · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part d only.

Question 9 continued

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(Total for Question 9 is 10 marks)

21. Question 9 · part d · 2 marks · Working and answer

Working and answer

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22. Question 9 · part c · 3 marks

2024 JUN · Paper 02

Further practice · Attempt part c only.

9

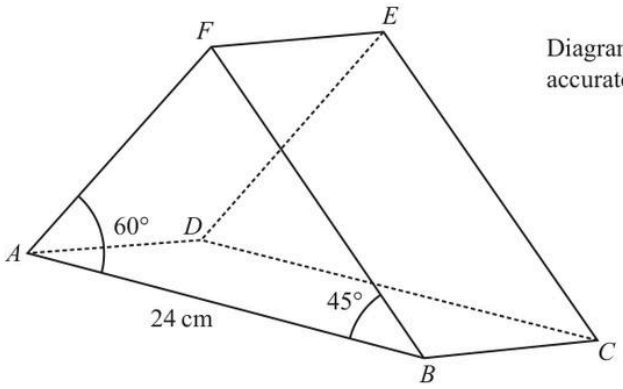


Figure 5

Figure 5 shows a right triangular prism $ABCDEF$ where $ABCD$ is a rectangle.

$AF = DE$ $BF = CE$ $AD = FE = BC$ $AB = DC = 24$ cm

$\angle ABF = \angle DCE = 45°$ $\angle BAF = \angle CDE = 60°$

Using a formula from page 2,

(a) show that $\sin AFB = \dfrac{\sqrt{2} + \sqrt{6}}{4}$

(3)

Without using a calculator,

(b) show that $BF = 12\left(3\sqrt{2} - \sqrt{6}\right)$ cm

(5)

The angle between the plane AEB and the plane $ABCD$ is 65°

(c) Find, in cm to 2 significant figures, the length of EF

(3)

(d) Find, in degrees to one decimal place, the size of the angle between the line CF and the plane $ABCD$

(4)

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22. Question 9 · part c · 3 marks · Working and answer

Working and answer

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